

Exploratory Data Analysis

27/05/26: Final Update

Data

The dataset has been provided by the Singapore Department of Statistics. It primarily covers births and fertility rates within Singapore, collecting data concerning a variety of variables from 1960 to 2025. However for the purpose of this data analysis the main variables that will be considered are:

- **Year:**
 - The EDA will be concerned with the entire period of 1960 to 2025. However for modelling and forecasting, it pertains to the periods 1960 to 2012, and 2013 to 2024 respectively.
- **TLF: Total Fertility Rate**
 - This refers to the average number of live-births each female would have during her reproductive years, subjected to prevailing age-specific fertility rates in the population in the given year.
 - Additionally data prior to 1980 pertains to total population while onwards pertains to resident population (Singaporean citizens and permanent residents).
- **TLB: Total Live Births**
 - Data is concerned with all live births within Singapore and its territorial waters as registered by the Singaporean government at the year of birth.
- **TLB_s: Total Live Births scaled**
 - This is an additional variable introduced. It is a down scaling of the live births data that will be used for ease of analysis.
 - E.g. If TLB were recorded 10,000 at the time then it's down scale would be 1.

Summary Statistics

Table 1: The first 6 rows of the dataset for display

Year	TFR	TLB	TLB_s
1960	5.76	61775	6.1775
1961	5.41	59930	5.9930
1962	5.21	58977	5.8977
1963	5.16	59530	5.9530
1964	4.97	58217	5.8217
1965	4.66	55725	5.5725

- Total Fertility rate on average is 1.52 with with a standard deviation of 1.23. This is relatively high compared to the mean. So fertility rate increases increases and decreases per year by this amount.

Table 2: Summary statistics of the Total Fertility Rate

Mean	Variance	Standard Deviation	0%	25%	50%	75%	100%
2.04	1.52	1.23	0.87	1.26	1.62	2.04	5.76

- On average the total live births per year is 4.41 or 44,100 with a standard deviation of 0.681. This standard deviation is lower than the mean. Total live births each year are increased and decreased by this amount or by 6,810.

Table 3: Sumamry statistics of the scaled Scaled Total Live Births

Mean	Variance	Standard Deviation	0%	25%	50%	75%	100%
4.41	0.464	0.681	3	3.95	4.24	4.85	6.18

Box-plots

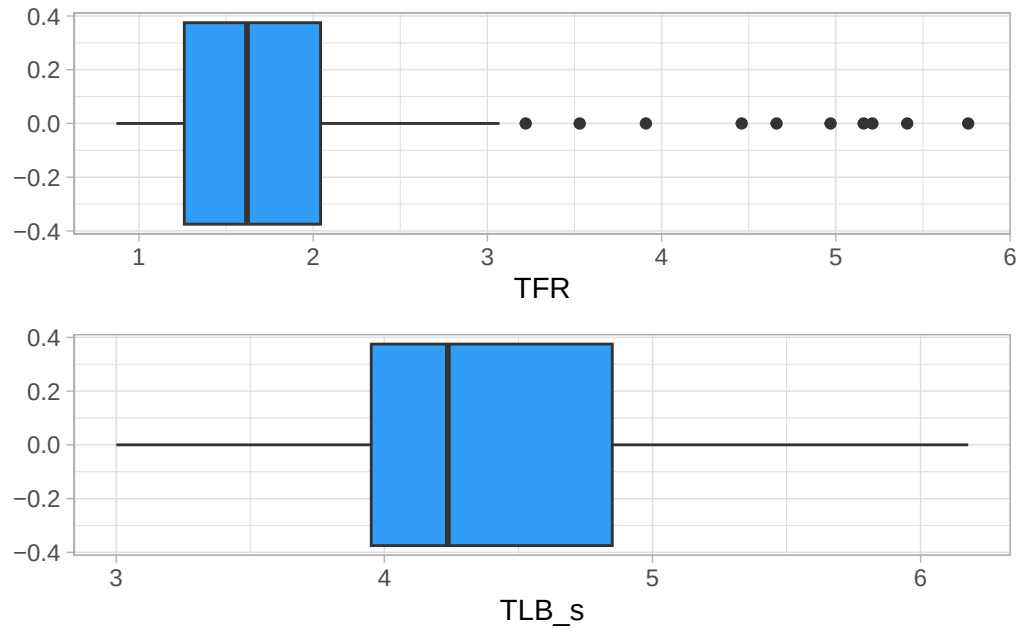


Figure 1: Box-plots of Total Fertility Rate (TFR) and Scaled Total Live Births (TLB_s)

The box-plot Total Fertility Rate has no skewed with 10 outliers. Suggesting overall lower fertility rate is recorded. 50% of recorded fertility rates lies roughly between 1.5 and 2. The lowest fertility rate is 0.87. Maximum fertility rate is 5.76 as all seen in Table 2. This is maximum is an outlier and supported by the box-plot. In addition about half the recorded fertility rates lie below or above the median; 1.62. This is also seen on fertility rate box-plot.

Total Live Births box-plot is right skewed with no outliers. 50% of recorded total live births lie roughly between 4 and the low 5s. The recorded median is 4.24 seen in Table 3. Half lie below and above this median. The maximums and minimums here aren't outliers but are recorded as 6.18 and 3 respectively in Table 2. Essentially at a one point in time, the maximum total live births was 61,800 and the minimum was 30,000.

Histograms

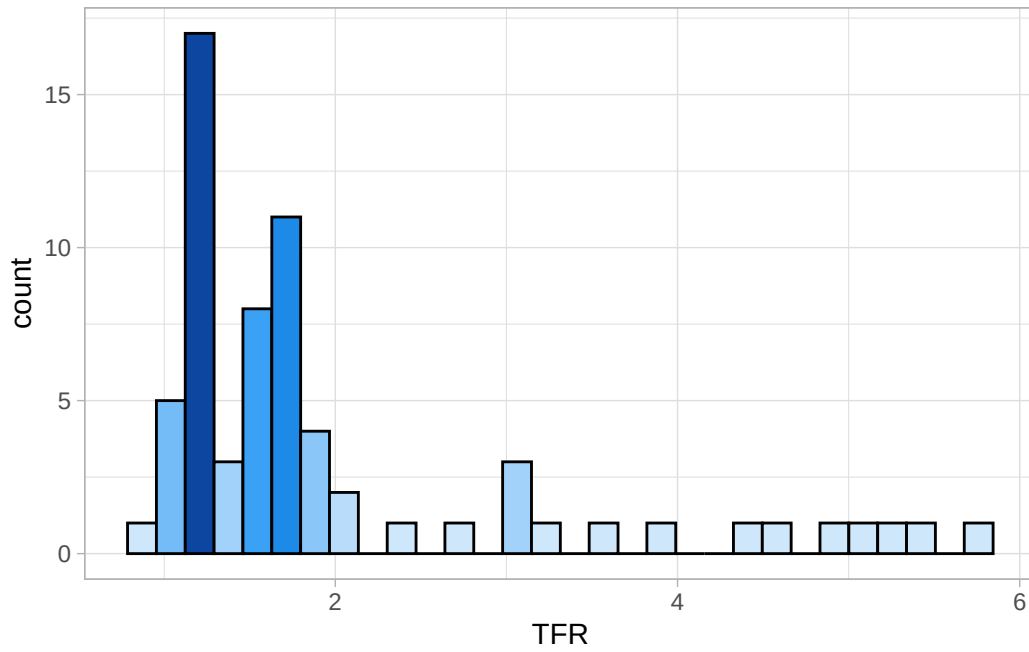


Figure 2: Histogram of Total Fertility Rate (TFR)

Histogram of the Total Fertility Rate is unimodal right-skewed. A majority of recorded data leans towards the lower end of fertility rate. As reflected by the box-plot there are also clusters of outliers here beyond the TFR of 3.

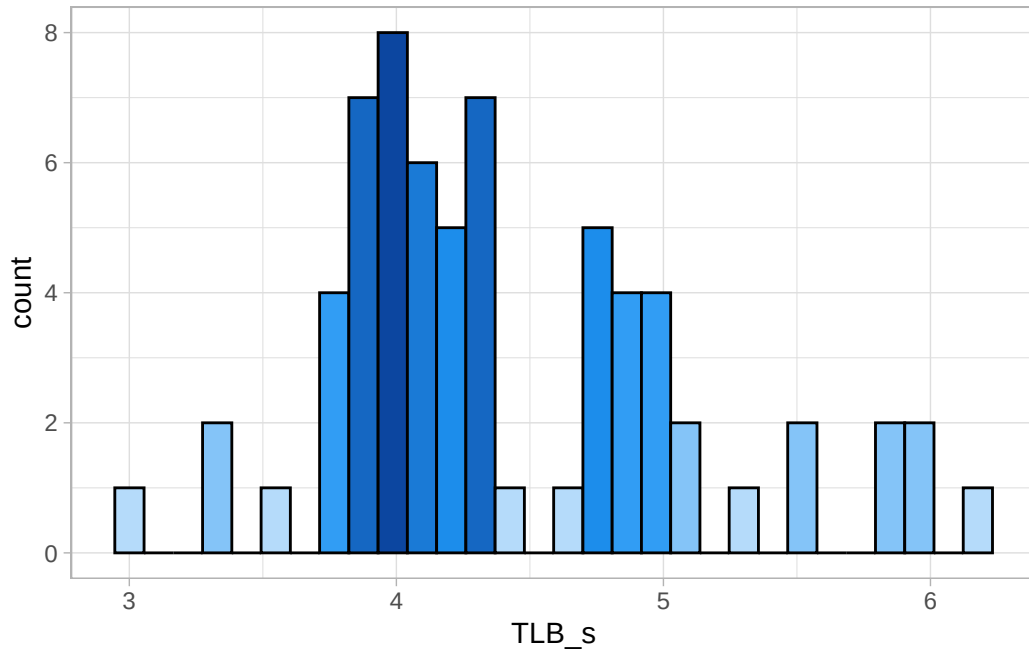


Figure 3: Histogram of scaled Total Live Births (TLB_s)

The histogram of Total Live Births displays a bimodal distribution. There's two main peaks are 4 and 5. Additionally there no data in the centre between 4 and 5 separating the two peaks.

Time Plots and Correlograms

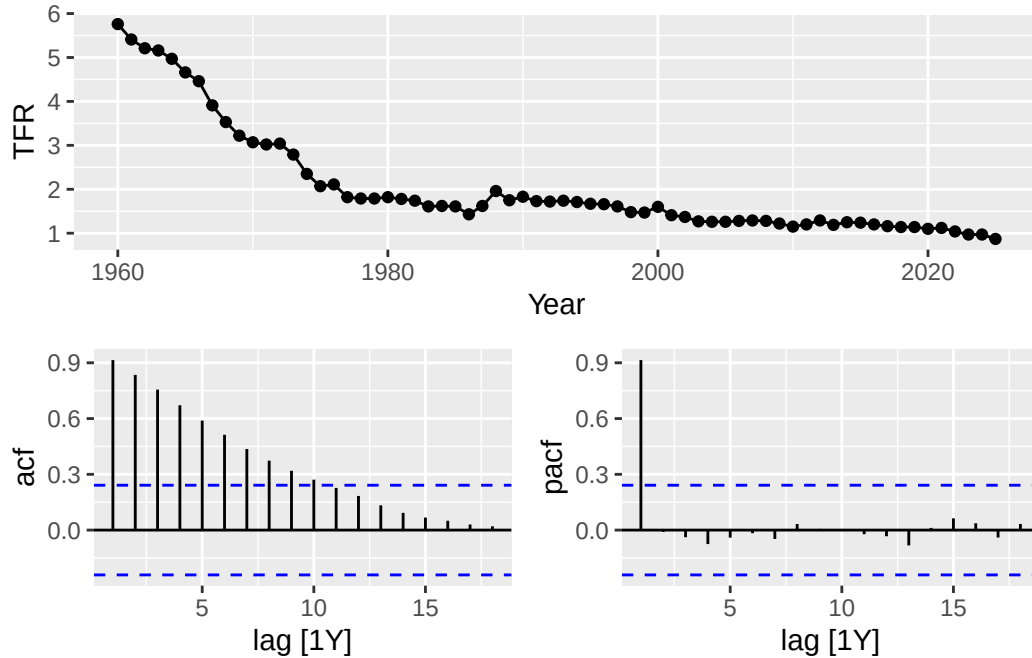


Figure 4: Time plot and Correlograms of Total Fertility Rate (TFR)

The time series for total fertility rate, its time plot has a decreasing trend. As it's annual data there is no seasonality. Its correlogram has a gradual decay. Past lag 10 the ACF is no longer significant. The partial correlogram cuts off sharply after lag 1. No more lags after have a significant PACF. Overall this suggests an autoregressive model could be considered. However, because of its non-stationarity and decreasing trend this isn't a definitive conclusion. Since there's a possibility of the trend influencing these correlogram. Transformation and differencing should be considered to ensure stationarity.

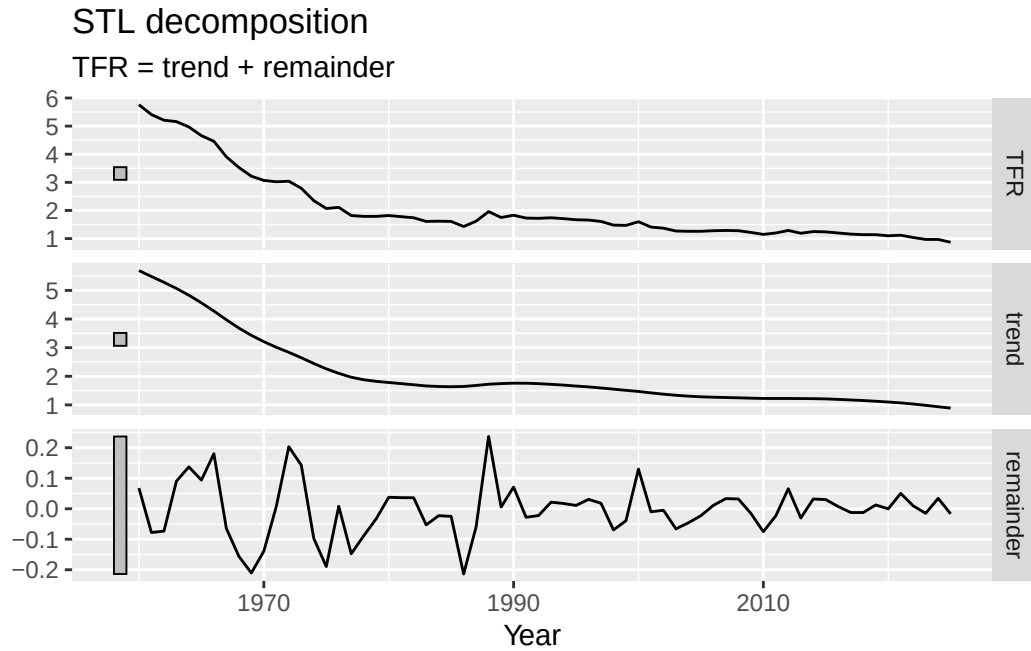


Figure 5: Time Series Decomposition of Total Fertility Rate (TFR)

The time decomposition of the total fertility rate further supports that it's a decreasing trend with no stationarity. The fluctuations seen in the remainder explains for the peaks and troughs seen in the time series. What these fluctuations could be is a potential avenue for further exploration. Additionally in the trend, it's clearly seen that it could be of an exponential decrease. So a logarithmic transform could be a consideration.

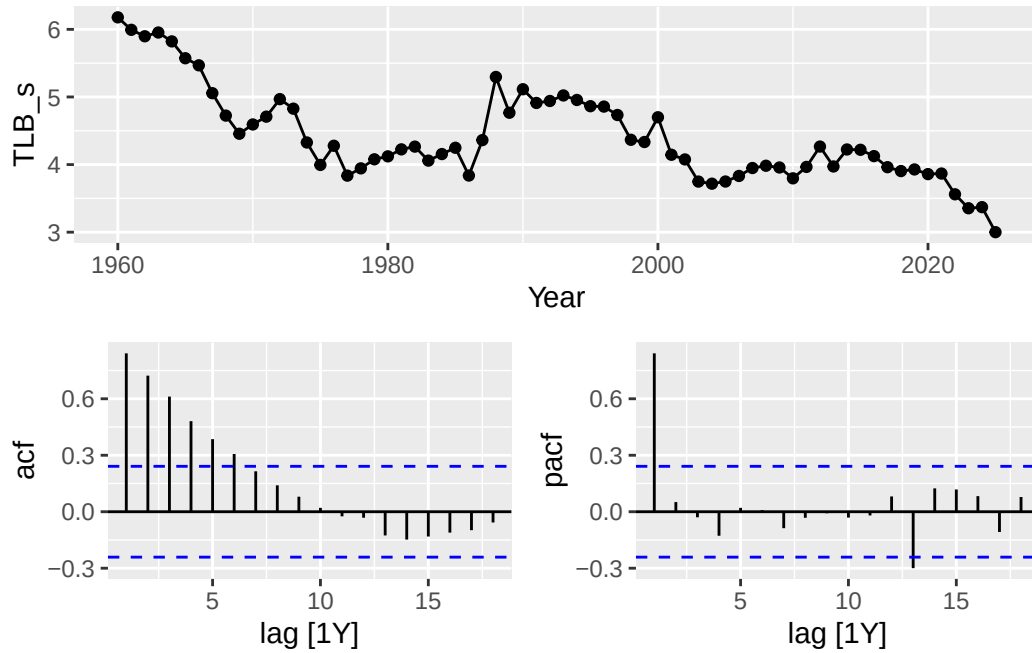


Figure 6: Time plot and Correlograms of scaled Total Fertility Rate (TLB)

Total Live Births, it's time plot displays a similar non-seasonal decreasing trend. In addition, both time plots display a peak increase in their respective variables, for the periods between the 1980s and 2000s. This could be an indication of socio-economical features not present in this data that could worth investigating. The correlogram features an ACF that decays gradually and is not significant after lag 6. It's partial correlogram, a significant peak in it's PACF and cuts off afterwards. Suggesting a similar consideration of autoregressive and a similar conclusion that the trend is influencing these.

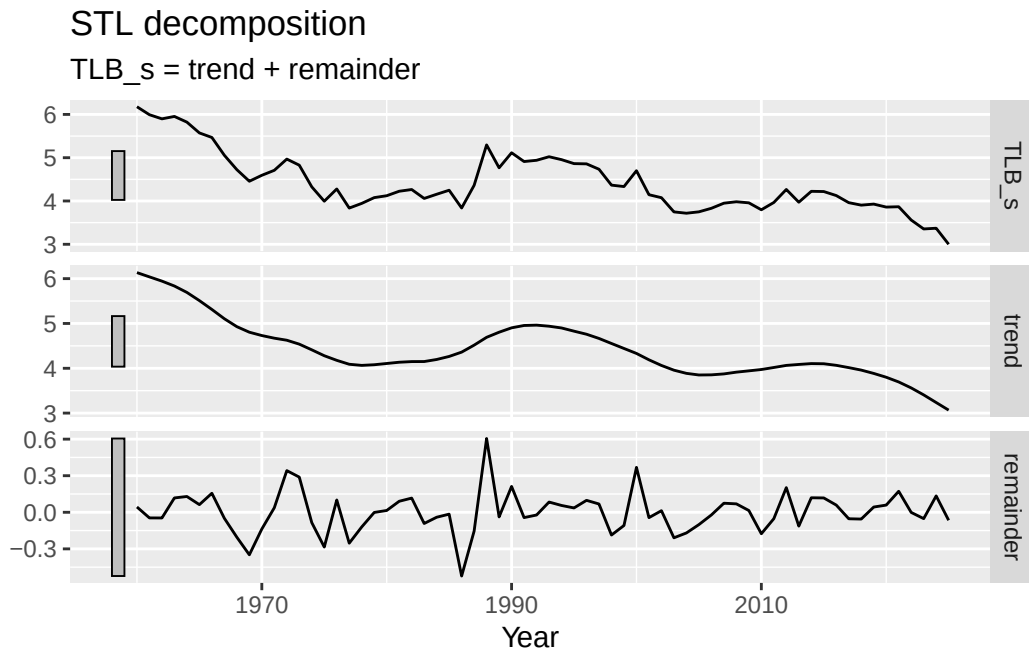


Figure 7: Time decomposition of scaled Total Live Births

The Total Live Births decomposition shows that the fluctuations seen in the time plot are explained by a remainder, leaving only a decreasing trend. This noticeable supports the the conclusion that between 1980 and 2000 there was an increase in live births bucking the trend.

Overall there's a correlation between fertility rate and live births. It's relationship could be worth considering if one is an significant indicator of the other.

Modelling

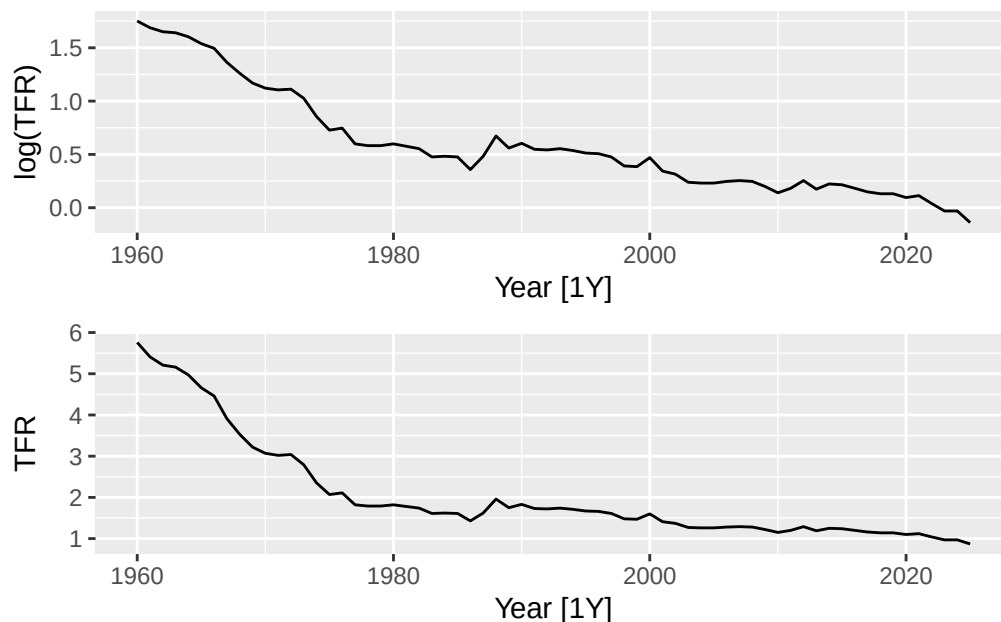


Figure 8: TFR vs it's logarithmic transformed. Marginal reduction in it's variation

A logarithmic transformation was considered in an attempt to make the Total Fertility Rate stationary and minimise it's variation. However as seen here, there is questionable marginal affect compared to the original data. So a logarithmic transformation is not in the final candidate models.

Table 4: The Unit Root Tests for Total Fertility Rate

KPSS Stat	p-value	Difference
1.07	0.01	2

A more formal way was conducted to determine if the time series for Total Fertility rate and Total Live Births are stationary, the Kwiatkowski-Phillips-Schmidt-Shin (KPSS) Test. The assumed null hypothesis was that the time series were stationary and reject this if it's p-value was less than 0.05. Additionally this test will also directly indicate how many differences is required to make it stationary. Hence a difference of order 2 is required as seen in Table 4.

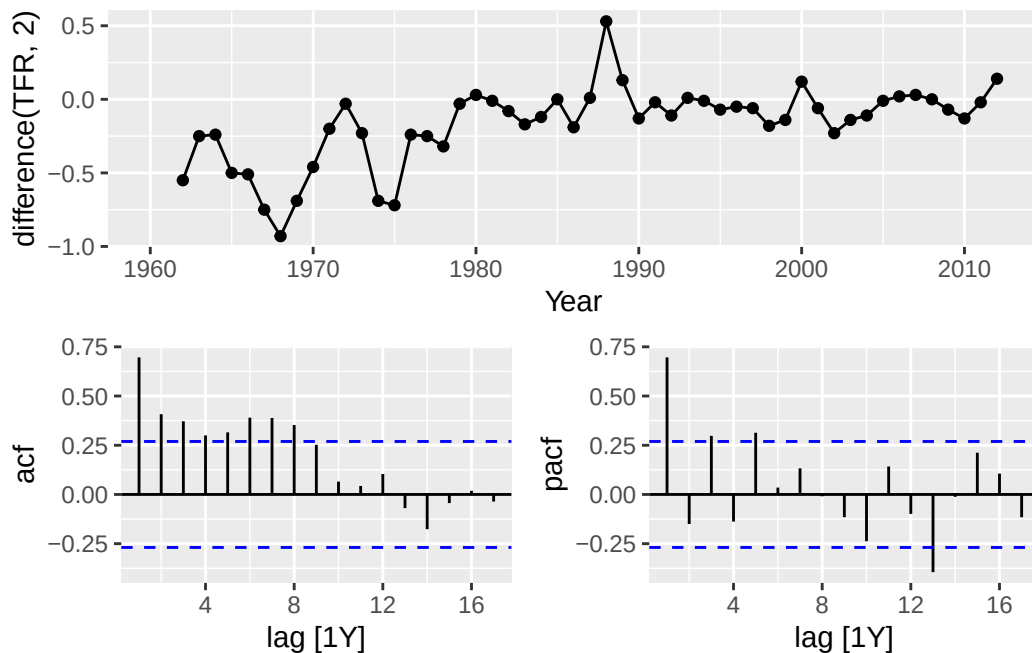


Figure 9: Twice differenced of the Total Fertility Rate (TFR)

The twice differenced time series now meets stationary assumptions. However it displays a minor increase trend between 1960 and 1980. Further differencing would mean overfitting. Hence alternative models may be considered if the candidate model isn't sufficient. The correlogram now has an ACF that gradually decays as supported by its original non-difference time series. The partial now has significant lags past 1, notably lags 3, 5, and a negative lag 12. There's no clean cut off as seen initially. Again suggesting an AR model.

Therefore can be concluded a candidate model for Total Fertility Rate could be an ARIMA(0,2,1) model as it's now also differenced.

Table 5: The Unit Root Tests for Total Live Births

KPSS Stat	p-value	Difference
0.624	0.0205	1

The same formal test for stationarity, the KPSS was conducted for the Total Live Births time series seen in Table 5. A difference of one was taken to ensure stationarity.

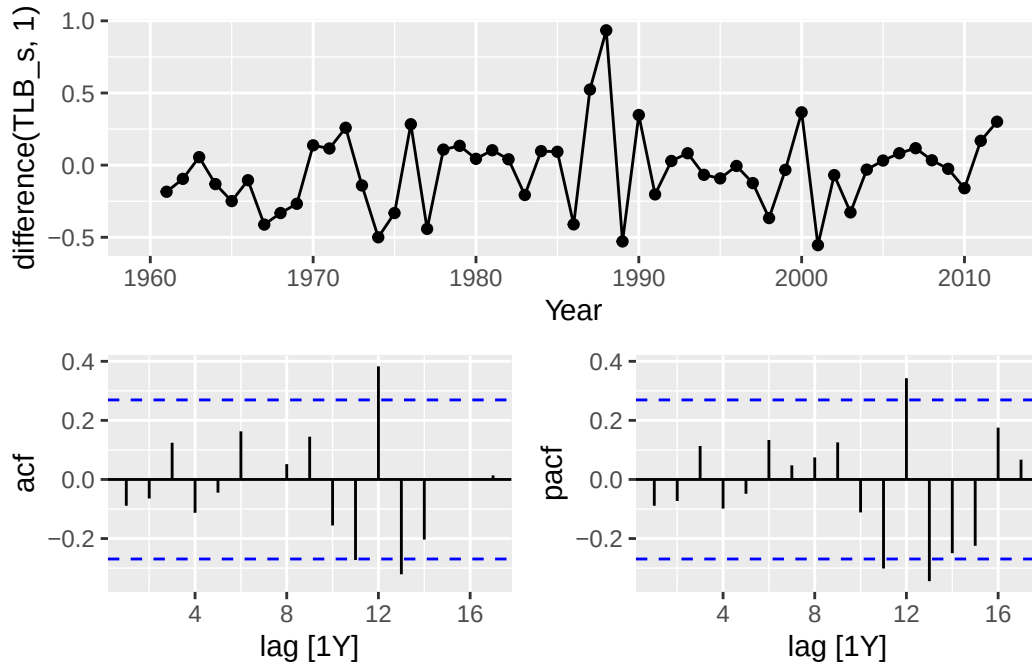


Figure 10: Once differenced scaled Total Live Births (TLB)

The time plot is now stationary however it's correlograms indicate no clear pattern to suggest either an AR or an MA model. A significant peak at lag 12 and similar negative lags at 11 and 13. This could be a random walk as it displays long periods of apparent trends up or down. Sudden and unpredictable changes in changes additionally. An indication of a stochastic trend.

For the Total Live Births, two models will be considered as candidate and for later analysis and validation. These are ARIMA(1,1,0) to compensate for the initial autoregressive indications seen earlier, and ARIMA(0,1,0) for the possible random walk.

Fitting

Table 6: Model Table for TFR and TLB

Models	Type	Estimate	Standard Error	Statistics	p-value
TFR_120	ar1	-0.384	0.128	-2.99	0.00432
TLB_010	constant	-0.0368	0.0381	-0.964	0.339
TLB_110	ar1	-0.0692	0.139	-0.497	0.621

For Total Fertility Rate an ARIMA(1,2,0) was fitted and for Total Live Births two models were fitted. An ARIMA(0,1,0) and ARIMA(1,1,0) seen in Table 7. From the table the models for fertility rate and live births are:

- ARIMA(1,2,0): $\Delta^2\text{TFR}_t = -0.384\Delta^2\text{TFR}_{t-1} + \epsilon_t$
 - Where Δ is the difference, back-shift operator.
- The random walk, ARIMA(0,1,0): $\Delta\text{TLB}_t = -0.0368 + \epsilon_t$
- ARIMA(1,1,0): $\Delta\text{TLB}_t = -0.0692 + \epsilon_t$

The p-values from Table 6 for the two Total Live Birth Models are greater than 0.05, so they're not statistically significant models. Alternative models should be considered and compared with these candidate models to see which is better at forecasting.

Models used for simple forecasting methods can be considered as benchmarks to evaluate if these candidate models are suitable for forecasting future data. These methods are the mean, naive, and drift methods. The BIC and AICc can be considered as well to determine the order of ARIMA models. Comparing these starting models with an optimised one can be useful. As good models tend to have minimised AIC, AICc or BIC.

Additionally their residuals and their fitted of these and optimised models can be compared and evaluated. Including residual diagnostics ranging from histograms correlograms, and Portmanteau tests (i.e. Box-Piece or Ljung-Box). Further checks of stationarity with an optimised model for Total Live Births for example can result in better forecasting compared to the candidate models.

Research Questions

1. How viable are these candidate models for forecasting the periods from 2013 to 2024 compared to more suitable optimised models?
2. How much can an optimised model explain the irregularities and fluctuations within both time series?
3. Given historical context, through industrialisation and government interventionist policies, Singapore saw rapid economic growth between the 1950s and 1990s. Becoming one of Four Asian Tigers predominantly as a financial hub. Additionally COVID-19 saw economic down turns, quarantines and border closures. How much of these fluctuations seen in these time series (notably the increases in the 20th century seen in the data) can be explained by these periods of significant progress and upheaval?
 - Main limitation of this is the current data. There isn't enough, a large enough variety such as GDP, total population growth, and other metrics concerning socio-economic factors to further investigate.
 - Additionally it may be beyond the scope of this analysis, as is eventually is a rabbit hole of other questions encapsulating various other fields:
 - Whether some increases in fertility rate due to early economic prosperity affected ethnic and age groups differently?
 - How much of this economic growth reversed and decreased Total Live Births and Total Fertility Rates? For example, developed countries typically see a down turn in reproduction and population growth. Be it due to work life balances and generational culture changes that make it uneconomical to reproduce.

Statistical Appendix

GitHub for the data cleaning and EDA coding: <https://github.com/Simon-a1740071/MATHS-X313-Report>

ARIMA Model

This is a combination of autoregression and moving average, where I in ARIMA is integration as in the reverse of differencing. A typical full ARIMA(p,d,q) model is written as:

$$y'_t = c + \sum_{i=1}^p \phi_i y'_{t-i} + \sum_{i=1}^q \theta_i \epsilon_{t-i}$$

- y'_t is the differenced series

Autocorrelation

For lag k , the ACF is the measure between the relationship between y_t and y_{t-k} :

$$\rho(k) = r_k = \frac{\sum_{t=k+1}^T (y_t - \bar{y})(y_{t-k} - \bar{y})}{\sum_{t=1}^T (y_t - \bar{y})^2}$$

Partial Autocorrelation

The theoretical partial autocorrelation function of a stationary time series is calculated using the Durbin-Levinson Algorithm

$$\phi_{n,n} = \frac{\rho(n) - \sum_{k=1}^{n-1} \phi_{n-1,k} \rho(n-k)}{1 - \sum_{k=1}^{n-1} \phi_{n-1,k} \rho(k)}$$

Where

- $\phi_{n,k} = \phi_{n-1,k} - \phi_{n,n} \phi_{n-1,n-k}$ for $k \in [1, n-1]$
- $\rho(\cdot)$ is the autocorrelation function

References

- Singapore Department of Statistics. (2026). *Births and fertility rates, annual* [Data set]. data.gov.sg. https://data.gov.sg/datasets/d_e39eeaeadb571c0d0725ef1eec48d166/view
- Singapore Department of Statistics. (2026). *Births, crude birth rate (CBR), total fertility rate (TFR) and reproduction rates* [Data set]. SingStat Table Builder. <https://tablebuilder.singstat.gov.sg/table/TS/M810091>